

Daniel Carranza

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EDUCATION

- 2023–present Ph.D. in Mathematics, Johns Hopkins University
Expected 2029
- 2022–2023 Program Associate, SLMath (formerly MSRI)
- 2021–2022 M.Sc. in Mathematics, University of Toronto
- 2017–2021 B.Sc., Western University
Mathematics (Honours Specialization); Computer Science (Minor)

PUBLICATIONS

1. *Autoformalizing the calculation of $\pi_3(S^2)$* , with C. Liu, E. Riehl, and E. Rijke, preprint, 2026. [\[arXiv\]](#)
2. *Discrete homotopy hypothesis for n -types*, with C. Kapulkin, submitted, 2026. [\[arXiv\]](#)
3. *Cubical models of ∞ -presheaves and the Bousfield–Kan formula*, with K. Arakawa and C. Kapulkin, *Adv. Math.* **503** (2026), Paper No. 111217. [\[arXiv\]](#) [\[Adv. Math.\]](#)
4. *Derived mapping spaces of ∞ -categories*, with K. Arakawa and C. Kapulkin, *Algebr. Geom. Topol.* (to appear), 2025. [\[arXiv\]](#)
5. *Categorical foundations of discrete dynamical systems*, with C. Kapulkin, N. Kershaw, R. Laubenbacher, and M. Wheeler, submitted, 2025. [\[arXiv\]](#)
6. *Nonexistence of colimits in naive discrete homotopy theory*, with C. Kapulkin and J. Kim, *Appl. Categ. Structures* **31** (2023), no. 5, 41, 6 pp. [\[arXiv\]](#) [\[ACS\]](#)
7. *Calculus of fractions for quasicategories*, with C. Kapulkin and Z. Lindsey, submitted, 2023. [\[arXiv\]](#)
8. *Diagonal lemma for presheaves on Eilenberg–Zilber categories*, with C. Kapulkin and L.-Z. Wong, *Theory Appl. Categ.* **44** (2025), Paper No. 11, 326–343. [\[arXiv\]](#) [\[TAC\]](#)
9. *Cofibration category of digraphs for path homology*, with B. Doherty, C. Kapulkin, M. Opie, M. Sarazola, and L.-Z. Wong, *Algebr. Comb.* **7** (2024), no. 2, 475–514. [\[arXiv\]](#) [\[ALCO\]](#)
10. *The Hurewicz theorem for cubical homology*, with C. Kapulkin and A. Tonks, *Math. Z.* **305**, 61 (2023), 20 pp. [\[arXiv\]](#) [\[Math. Z.\]](#)
11. *Cubical setting for discrete homotopy theory, revisited*, with C. Kapulkin, *Compos. Math.* **160** (2024), no. 12, 2856–2903. [\[arXiv\]](#) [\[Compos. Math.\]](#)
12. *Homotopy groups of cubical sets*, with C. Kapulkin, *Expo. Math.* **41** (2023), no. 4, 125518, 55 pp. [\[arXiv\]](#) [\[Expo. Math.\]](#)
13. *2-adjoint equivalences in homotopy type theory*, with J. Chang, C. Kapulkin, and R. Sandford, *Log. Methods Comput. Sci.* **17** (2021), no. 1, Paper No. 3, 9 pp. [\[arXiv\]](#) [\[LMCS\]](#)

TALKS

The discrete homotopy hypothesis for n -types

- September 2026 [Homology and Homotopy in the World of Graphs](#), Bernoulli Center
- July 2026 [Category Theory 2026](#), Baltimore, Maryland
- April 2026 Discrete homotopy theory event, University of Western Ontario

Generalizing the Bousfield–Kan formula

December 2025 [Canadian Mathematical Society Winter Meeting](#), Toronto, Ontario

Weak homotopy types of finite spaces

July 2025 [Mathematical Congress of the Americas](#), Miami, Florida

June 2025 [Canadian Mathematical Society Summer Meeting](#), Quebec City, Quebec

Nonexistence of colimits in naive discrete homotopy theory

November 2024 [Topology and Geometry seminar](#), University of Florida

January 2024 [Joint Mathematics Meetings](#), San Francisco, California [\[slides\]](#)

Calculus of fractions for quasicategories

February 2025 [Topology Seminar](#), University of Virginia

March 2024 [AMS Spring Southeastern Sectional Meeting](#), Florida State University

December 2023 [Canadian Mathematical Society Winter Meeting](#), Montreal, Quebec

October 2023 [Geometry, Topology and Physics seminar](#) (online), NYU Abu Dhabi

March 2023 [Algebraic Topology seminar](#), UCLA

Hurewicz theorem in discrete homotopy theory

November 2023 [9th Mexican Workshop on Applied Geometry and Topology](#) (online), Centro de Investigación en Matemáticas (CIMAT)

Cubical setting for discrete homotopy theory

December 2023 [Geometric, Algebraic, and Topological Combinatorics](#), Mathematisches Forschungsinstitut Oberwolfach

April 2023 [Math & Statistics Colloquium](#), Reed College

April 2023 [Algebraic & Geometric Topology Seminar](#), University of Nevada, Reno

March 2023 [AIM Workshop: Discrete and Combinatorial Homotopy Theory](#)

November 2022 [Berkeley Seminar](#), Topos Institute [\[video\]](#)

October 2022 [AMS Fall Eastern Sectional Meeting](#), University of Massachusetts Amherst

May 2022 [Ontario Combinatorics Workshop](#), University of Waterloo [\[slides\]](#)

Formalizing 2-adjoint equivalences in homotopy type theory

August 2020 [Western Homotopy Theory seminar](#), University of Western Ontario [\[video\]](#)

Formal verification of mathematics

August 2020 [Canadian Undergraduate Mathematics Conference](#) (online) [\[video\]](#)

SCHOLARSHIPS AND AWARDS

2024–2027 [Canada Graduate Scholarship – Doctoral \(CGS D\)](#)

Summer 2021 [NSERC Undergraduate Student Research Award](#), Western University
Cubical methods in discrete homotopy theory

Summer 2020 [NSERC Undergraduate Student Research Award](#), Western University
2-adjoint equivalences in homotopy type theory

2018–2021 [Dean’s Honor List](#), Western University

September 2017 [Western Scholarship of Excellence](#)

TEACHING EXPERIENCE

Teaching assistant, University of Toronto

- Winter 2022 Math Learning Centre
- Winter 2022 Engineering Calculus 2
- Fall 2021 Engineering Linear Algebra

OTHER ACTIVITIES

- Summer 2025 Fields Undergraduate Summer Research Program, Western University
Co-supervisor, Project 09: Formal verification of topological combinatorics
- December 2023 Oberwolfach Workshop 2350: Geometric, Algebraic, and Topological Combinatorics, Oberwolfach Research Institute for Mathematics
Workshop participant
- March 2023 AIM Workshop: Discrete and Combinatorial Homotopy Theory, San Jose, California
Workshop participant
- June 2022 Higher Categories and Categorification, Part Two, MSRI
Program participant
- February 2020 Higher Categories and Categorification, MSRI
Research visit